

Experiment 6: Sketch Sequence diagram for the project

Learning Objective: Students will be able to draw Sequence diagrams for the project.

Tools: Dia, StarUML

Theory: A sequence diagram shows, as parallel vertical lines (lifelines), different processes or objects that live simultaneously, and, as horizontal arrows, the messages exchanged between them, in the order in which they occur. This allows the specification of simple runtime scenarios in a graphical manner.

Sequence Diagram representation

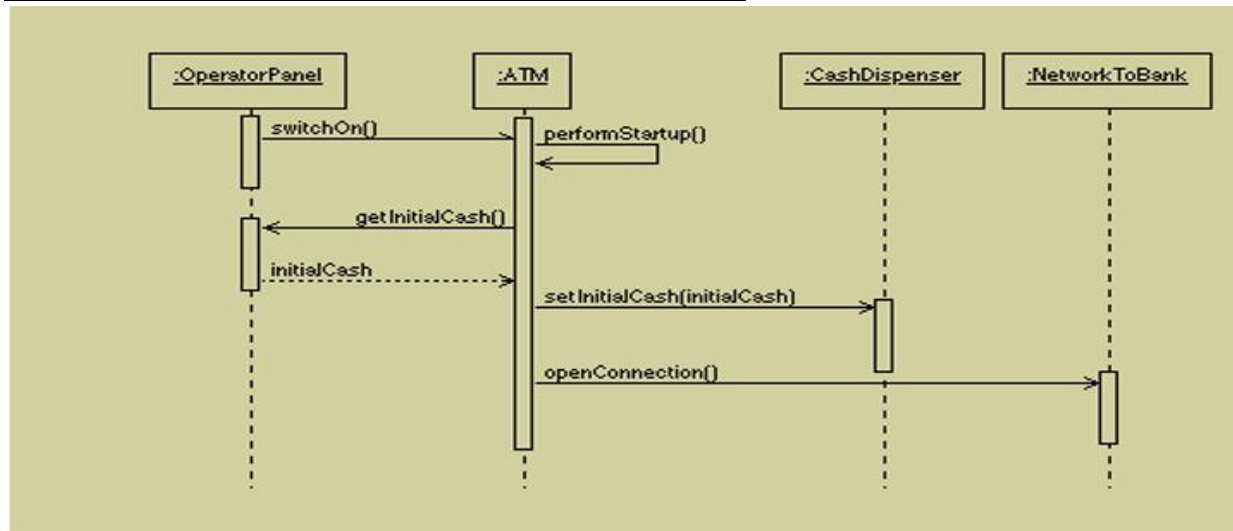
Call Message: A message defines a particular communication between Lifelines of an Interaction.

Destroy Message: Destroy message is a kind of message that represents the request of destroying the lifecycle of the target lifeline.

Lifeline: A lifeline represents an individual participant in the Interaction.

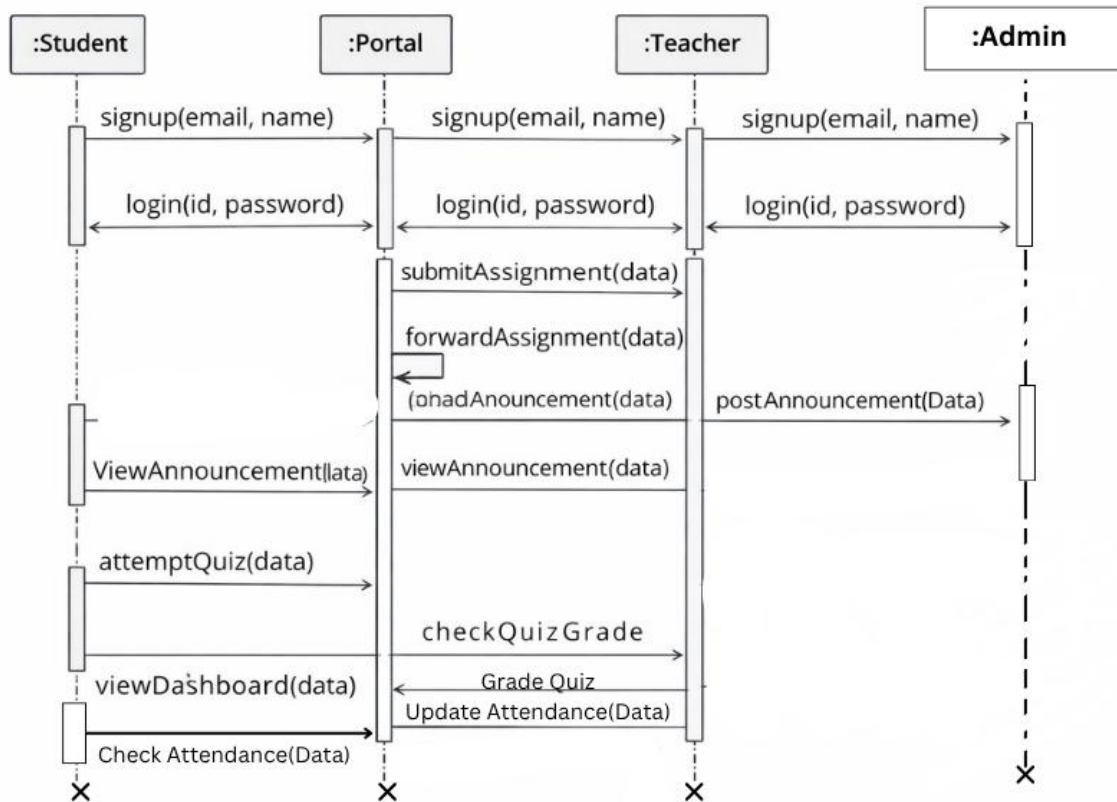
Recursive Message: Recursive message is a kind of message that represents the invocation of a message of the same lifeline. Its target points to an activation on top of the activation where the message was invoked from.

Sequence Diagram: Example for ATM System startup



It is clear that sequence charts have a number of very powerful advantages. They clearly depict the sequence of events, show when objects are created and destroyed, are excellent at depicting concurrent operations, and are invaluable for hunting down race conditions. However, with all their advantages, they are not perfect tools. They take up a lot of space, and do not present the interrelationships between the collaborating objects very well.

Result and Discussion:



Learning Outcomes: Students should have the ability to

LO1: Identify the classes and objects.

LO2: Identify the interactions between the objects

LO3: Develop a sequence diagram for different scenarios

Outcomes: Upon completion of the course students will be able to draw the sequence diagram for the project.

Conclusion:

1. What is a sequence diagram?

ans) A sequence diagram is a UML diagram that shows how objects interact with each other in a time-ordered sequence of messages.

2. What are advantages of sequence diagram?

- Clarifies object interactions step by step.
- Helps in visualizing system behavior.
- Useful for validating logic before coding.

3. What are entities in sequence diagram?

ans) Entities are the objects, actors, and lifelines that participate in the interaction, exchanging messages over time.

4. Explain its relation with the class diagram.

ans) A sequence diagram shows dynamic interactions between objects, while a class diagram shows their static structure and relationships.

Correction Parameters	Formative Assessment [40%]	Timely completion of Practical [40%]	Attendance / Learning Attitude [20%]	
Marks Obtained				